

Predictive Analytics Using Historical Data

Overview

This project builds a predictive model using historical monthly sales data to forecast future sales trends. The assignment requires data cleaning, model building, evaluation, and visualization of predictions.[cite:1]

Dataset

The dataset used in this project contains two columns: `Month` and `Sales`. It represents monthly historical sales values and is suitable for simple trend forecasting using a time-based regression approach.

Data Preprocessing

The historical data was cleaned by converting the `Month` column into a date format, sorting the records in chronological order, and handling missing values where necessary. A numeric `time_index` feature was then created so the model could learn the sales trend over time.[cite:1]

Model Development

A linear regression model was used to predict sales based on the time index. The dataset was split into training and testing sets using chronological order so that earlier months were used for training and later months were used for evaluation.[cite:1]

Evaluation

Model accuracy was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). These metrics help measure the gap between actual sales and predicted sales and indicate whether the model captures the underlying sales trend effectively.[cite:1]

Visualizations

The notebook includes visualizations for historical sales trends, actual versus predicted sales values, and future sales forecasts. These charts make it easier to interpret the model's performance and understand the projected direction of future sales.[cite:1]

Conclusion

The predictive analytics workflow demonstrates how historical sales data can be used to forecast future business trends. Even with a simple linear regression model, the project shows

how data preprocessing, model evaluation, and visualization can support basic data-driven decision-making.[cite:1]